

**CENTER FOR DRUG EVALUATION AND
RESEARCH**

APPLICATION NUMBER:

761248Orig1s000

**RISK ASSESSMENT and RISK MITIGATION
REVIEW(S)**

Division of Risk Management (DRM)
Office of Medication Error Prevention and Risk Management (OMEPRM)
Office of Surveillance and Epidemiology (OSE)
Center for Drug Evaluation and Research (CDER)

Application Type	BLA
Application Number	761248
PDUFA Goal Date	March 12, 2024
Nexus TTT #	2022-2424
Reviewer Name(s)	Carla Darling, Pharm.D., BCPS
Team Leader	Jacqueline Sheppard, Pharm.D.
Division Director	Cynthia LaCivita, Pharm.D.
Review Completion Date	June 28, 2024
Subject	Evaluation of Need for a REMS
Established Name	Donanemab-azbt
Trade Name	Kisunla
Name of Applicant	Eli Lilly and Company
Therapeutic Class	Anti-Amyloid Beta Monoclonal Antibody
Formulation(s)	Solution for Injection
Dosing Regimen	700 mg administered as an intravenous infusion every four weeks for the first three doses, followed by 1400 mg every four weeks

Table of Contents

EXECUTIVE SUMMARY	3
1. Introduction	4
2. Background.....	4
2.1. Product Information.....	4
2.2. Regulatory History.....	5
3. Therapeutic Context and Treatment Options.....	5
3.1. Description of the Medical Condition.....	5
3.2. Description of Current Treatment Options.....	6
4. Benefit Assessment	6
5. Risk Assessment & Safe-Use Conditions.....	8
5.1. Serious Adverse Events (SAE).....	9
5.2. Adverse Events of Special Interest (AESI)	9
6. Expected Postmarket Use.....	14
7. Risk Management Activities Proposed by the Applicant.....	14
8. Discussion of Need for a REMS.....	14
9. Conclusion & Recommendations.....	16
10. Appendices.....	16
10.1. References	16
10.2. Table 1. Drugs Approved in the US for Alzheimer's Disease	18

EXECUTIVE SUMMARY

This review by the Division of Risk Management (DRM) evaluates whether a risk evaluation and mitigation strategy (REMS) for the new molecular entity Kisunla (donanemab-azbt) is necessary to ensure the benefits outweigh its risks. On June 12, 2023, Eli Lilly and Company (Applicant) re-submitted a Biologic Licensing Application (BLA) 761248 for donanemab-azbt, an amyloid beta (A β) directed antibody with the proposed indication [REDACTED] (b) (4) of Alzheimer's disease (AD). During the review of the application, the indication was revised to the treatment of Alzheimer's disease. The Applicant did not submit a proposed REMS or risk management plan with this application.

Alzheimer's Disease (AD) is a progressive, degenerative neurologic disorder characterized by progressive memory loss, behavioral problems, and inability to perform activities of daily living. AD is the most common cause of dementia and a leading cause of morbidity and mortality in the aging population. As there are no approved treatments designed to cure the disease, there remains an unmet need for treatment options for AD.

The efficacy of donanemab-azbt was evaluated in one pivotal phase 3 trial (Study AACI). The clinical efficacy reviewer determined that based on this trial, the Applicant provided substantial evidence of effectiveness and demonstrated a clinically meaningful and statistical reduction in clinical decline. The risks associated with donanemab-azbt include hypersensitivity and infusion related reactions as well as amyloid related imaging abnormalities (ARIA), including ARIA-edema (ARIA-E) and ARIA-hemosiderin deposition (ARIA-H).

Based on the safety and efficacy information available, DRM and the Division of Neurology 1 (DN1) have determined that a REMS is not needed to ensure the benefits of donanemab-azbt outweigh its risks. The risk of ARIA, particularly in patients who are ApoE ϵ 4 homozygotes, and the risk of intracerebral hemorrhage will be included in labeling in a boxed warning. The boxed warning will also describe fatal intracerebral hemorrhages that occurred in subjects treated with donanemab-azbt, and caution physicians to consider whether focal neurologic symptoms could be due to ARIA-E when considering thrombolytic therapy in a patient taking donanemab-azbt. Labeling will convey recommendations for MRI monitoring, radiographic classification criteria for ARIA severity, the need for assessment of symptoms associated with ARIA throughout treatment, considerations for continuing donanemab-azbt in the setting of ARIA, and the need for ApoE ϵ 4 status prior to initiation of treatment. The Applicant also intends to provide a safety card for patients that includes ARIA-related information including details to inform and aid healthcare providers in managing emergency situations. The risks of hypersensitivity and infusion related reactions will be included in the Warnings and Precautions section of labeling. Other approved anti-amyloid beta monoclonal antibodies, Aduhelm (aducanumab-avwa) by accelerated approval and Leqembi (lecanemab-irmb) by traditional approval, share similar risks so prescribers are likely to be aware of the risks and the need for monitoring and management. Neither Aduhelm nor Leqembi are subject to a REMS. In addition, a Medication Guide will communicate the risks to patients and caregivers.

At the time of this review, final labeling, enhanced post-market pharmacovigilance and post-marketing requirements (PMRs) were still under review. If new safety information becomes available, DRM can re-evaluate the need for a REMS.

1. Introduction

This review by the Division of Risk Management (DRM) evaluates whether a risk evaluation and mitigation strategy (REMS) for the new molecular entity (NME),^a Kisunla (donanemab-azbt, hereafter referred to as donanemab) is necessary to ensure the benefits outweigh its risks. On May 18, 2022, Eli Lilly and Company (hereafter referred to as the Applicant) submitted the final portion of a rolling submission of a Biologic Licensing Application (BLA) 761248 for donanemab for accelerated approval with the proposed indication (b) (4) of Alzheimer's disease (AD).^{1,2} This Application is under review in Division of Neurology 1 (DN1). On January 18, 2022, BLA 761248 received a complete response letter from the DN1, based on a safety database that was insufficient to characterize the long-term safety of donanemab.³ The resubmission of BLA 761248 received on June 12, 2023, contains data from the Phase 3 Study AACI.⁴ The Applicant did not submit a proposed REMS or risk management plan with this application.

2. Background

2.1. Product Information

Donanemab is a humanized immunoglobulin gamma 1 (IgG1) monoclonal antibody (MAb) directed against insoluble truncated pyroglutamate amyloid beta.⁵ A defining pathophysiology of AD includes accumulation of amyloid beta plaques in the brain. Donanemab reduces amyloid plaques in the brain of people with AD.¹ It is theorized that donanemab's effect on amyloid plaques will attenuate the pathological course of AD and thereby slow disease progression.⁶

Donanemab is proposed as an intravenous infusion of 700 mg administered every four weeks for the first three doses, followed by 1400 mg every four weeks.⁵ Donanemab is not currently approved in any jurisdiction.⁷ The continuation, reduction or cessation of donanemab dosing in clinical trials was dependent on the findings of amyloid imaging including amyloid positron emission tomography (PET) levels and for treatment-emergent ARIA or symptoms that then showed ARIA-E or ARIA-H on MRI.^b Donanemab will most likely be administered within health care settings, such as infusion centers and provider clinics/offices. Donanemab was granted break through therapy designation, fast track designation, and priority review. If approved, donanemab would be the second anti-amyloid antibody with traditional approval for the treatment of AD.

^a Section 505-1 (a) of the FD&C Act: *FDAAA factor (F): Whether the drug is a new molecular entity.*

^b Section 505-1 (a) of the FD&C Act: *FDAAA factor (D): The expected or actual duration of treatment with the drug.*

2.2. Regulatory History

The following is a summary of the regulatory history for BLA 761248 relevant to this review:

- 07/20/2018: Fast track designation granted
- 06/21/2021: Breakthrough therapy designation granted
- 08/26/2021: Rolling review request granted
- 05/18/2022: Final portion of rolling original BLA received²
- 07/15/2022: Priority review request granted
- 11/21/2022: A Late-cycle meeting was held between the Agency and the Applicant via teleconference. The Agency informed the Applicant that no issues related to risk management have been identified to date for donanemab
- 01/18/2023: Complete Response (CR) letter sent to the Applicant due to safety data that was inadequate to characterize the long-term safety of donanemab
- 06/12/2023: Type 2 resubmission received⁴
- 10/11/2023: Applicant submitted additional clinical information on 07/31/2023, 09/05/2023, and 10/10/2023, which the Agency determined constitute a Major Amendment that extended the PDUFA goal date to March 12, 2024⁸⁻¹¹
- 06/10/2024: Peripheral and Central Nervous System (PCNS) Advisory Committee was convened to discuss whether the data from the Phase 3 Study AACI (TRAILBLAZER-ALZ 2) support a favorable benefit-risk assessment for the use of donanemab for the treatment of Alzheimer's disease (AD)¹²

3. Therapeutic Context and Treatment Options

3.1. Description of the Medical Condition

AD is a progressive, degenerative neurologic disorder characterized by progressive memory loss, behavioral problems, and inability to perform activities of daily living.¹³ AD is the most common cause of dementia and a leading cause of morbidity and mortality in the aging population.^{13,14} About 6.9 million people who are age 65 and older in the United States (US) suffer from AD.^{14,c} Without interventions to prevent or slow the disease, it has been projected that this number could double by 2050.¹⁴ AD is currently the seventh leading cause of death in the US.^{14,15} In a 15-year prospective cohort study of

^c Section 505-1 (a) of the FD&C Act: *FDAAA factor (A): The estimated size of the population likely to use the drug involved.*

patients age > 65 years or older, the mortality rate in patients with Alzheimer's disease (unadjusted for age or sex) was observed to be 133.8/1000 person- years.^{16, d}

The clinical course of the disease is a continuum of preclinical disease, mild cognitive impairment, and dementia.^{13,14} Early in the disease, changes may be undetectable in affected patients but progress to subtle problems with memory and thinking, and ultimately difficulties with memory, language, and problem-solving, which limit the individual's ability to perform everyday activities. Patients suffering from AD experience significant morbidity while living with the disease due to progressive memory loss, behavioral problems, and loss of independence due to inability to perform activities of daily living. The burden associated with AD is significant due to high direct medical costs as well as the unpaid time the family may spend caring for those suffering with the disease.¹⁴ Life expectancy may vary depending on several factors, however, the average survival is 4-8 years after a diagnosis.^{13,17}

3.2. Description of Current Treatment Options

There are currently no therapies that cure or slow the clinical decline of AD.¹⁸ Current treatment goals are aimed at maintaining quality of life, treatment of cognitive symptoms, and management of behavioral and psychological symptoms of dementia.¹⁴ Available FDA approved drugs are cholinesterase inhibitors (e.g. donepezil, rivastigmine, and galantamine) for the treatment of mild, moderate, and severe AD dementia; N-methyl-D-aspartate antagonist, memantine, for the treatment of moderate to severe AD dementia; aducanumab and lecanemab for the treatment of AD in patients with mild cognitive impairment or mild dementia.^{18,19} Cholinesterase inhibitors and memantine only provide symptomatic benefits and are less beneficial as the disease progresses.¹⁸

Two other anti-amyloid beta MAbs have been approved for the treatment of Alzheimer's disease, specifically patients with mild cognitive impairment or mild dementia stage of disease. Anti-amyloid beta MAb treatments target the underlying pathophysiological changes associated with AD and include Aduhelm, which received accelerated approval in 2021 followed by Leqembi, which received traditional approval in 2023.^{20,21} The serious risk associated with aducanumab and lecanemab is amyloid related imaging abnormalities (ARIA), including ARIA-edema (ARIA-E) and ARIA-hemosiderin deposition (ARIA-H). While the reduction of amyloid beta by current anti-amyloid beta MAbs show a reduction in clinical decline, there is still a need for effective therapies that can delay, halt, or reverse the pathophysiological process of Alzheimer's disease. See Table 1 in Appendix 10.2 for a list of currently approved FDA treatments.

4. Benefit Assessment

The primary source for evidence of efficacy of donanemab in AD was a phase 3 Study, IST-MC-AACI (Study AACI), also called TRAILBLAZER-ALZ2 (NCT04437511). Study AACI consisted of a randomized, multicenter, double-blind, placebo-controlled study of 1727 patients with early symptomatic

^d Section 505-1 (a) of the FD&C Act: FDAAA factor (B): *The seriousness of the disease or condition that is to be treated with the drug.*

Alzheimer's disease. Subjects were required to have evidence of brain amyloid and tau pathology on PET scans. The study included a screening period, a 76-week placebo -controlled treatment and a follow up period of 44 weeks. Patients were randomized to receive donanemab (n=853) or placebo (n=874) in a 1:1 ratio in the placebo-controlled period. Randomization was stratified by investigative site and tau pathology (low/medium or high on PET scan). The donanemab group received 700 mg every 4 weeks for the first 3 doses, followed by 1400 mg every 4 weeks with titration to placebo at weeks 24, 52, and 76 if amyloid PET levels reached less than 11 CLs at any 1 measure or 11 to less than 25 CLs for 2 consecutive measures.

The primary endpoint was change in the composite measure integrated Alzheimer's Disease Rating Scale (iADRS) from baseline score to 76 weeks. The iADRS is a simple linear combination of the ADAS-Cog 13 and ADCS-iADL with score ranges from 0 to 144 and with lower scores indicating greater disease severity. There were two analysis populations based on tau PET scan including a low/medium tau population and combined population of both low/medium tau and high tau levels. Results show a statistically significant slowing of AD progression in the donanemab group compared to placebo as measured by change from baseline in iADRS in the overall population (2.92 [-22.3%], $p < 0.001$) and the low/medium population (3.25 [-35.1%], $p < 0.001$) at 76 weeks. Per the clinical efficacy reviewer, the low/medium tau and the overall populations are highlighted because they are the two key populations for the prespecified analysis plan.

Additional secondary endpoints include change from baseline at 76 weeks in clinical dementia rating scale – sum of boxes (CDR-SB), Cognitive Subscale containing 13 tasks (ADAS-Cog13), instrumental subscore of the Alzheimer's Disease Cooperative Study – Activities of Daily Living scale (ADCS-iADL), Mini-Mental State Examination (MMSE). The DN1's recommended endpoint, CDR-SB, show a statistically significant reduction in clinical decline for donanemab compared to placebo of -0.67 [-36%], $p < 0.001$ in the low/medium tau population and -0.70 (-29%), $p < 0.001$ in the overall population. Statistically significant results favoring donanemab in the overall population were observed for the multiplicity-controlled secondary endpoints ADAS-Cog 13 (-1.33 [-20%], $p < 0.001$) and ADCS-iADL (1.70 [-28%], $p < 0.001$). The clinical efficacy reviewer concluded that the effect on CDR-SB in Study AACI represents a clinically meaningful reduction of clinical decline.

Secondary biomarker endpoints include change from baseline at 76 weeks for brain amyloid PET, brain tau PET, brain volume magnetic resonance imaging (MRI), and plasma levels of neurofilament light chain (NfL), glial fibrillary acidic protein (GFAP), and phosphorylated tau (p-tau 181 and p-tau 217). Secondary biomarker endpoints show a statistically significant reduction from baseline in brain amyloid as measured by PET and reported as CLs at Week 76 (-86.4, $p < 0.0001$) in the overall population. No statistically significant difference from baseline to week 76 was observed in brain tau levels between groups. At week 76, donanemab-treated patients had statistically significantly greater reduction in brain volume and greater increase in ventricular volume than placebo-treated patients. At week 76, donanemab-treated patients had a statistically significant greater reduction in plasma p-tau 217, p-tau 181, and GFAP compared to placebo.

Supportive information was provided from the phase 2 Study IST-MC-AACG (Study AACG), also called TRAILBLAZER-ALZ (NCT03367403), which was reviewed by the DN1 during the initial review cycle. The

clinical efficacy reviewer determined that the effect of donanemab on brain amyloid plaque demonstrated in Study AACG was robust and persuasive and that the estimates of the treatment effect across secondary endpoints are consistent with the reduction of clinical decline observed in Study AACI.²²

The clinical efficacy reviewer concluded that the use of donanemab in AD resulted in clinically meaningful reduction of clinical decline in Study AACI based on a statistically significant reduction of the DN1's recommended endpoint, CDR-SB.²² The DN1 determined that Applicant provided substantial evidence of effectiveness based on efficacy data from a single, adequate, and well-controlled large multicenter trial (Study AACI) that demonstrated a clinically benefit of donanemab in AD with support from a smaller study (AACG) that demonstrated similar effects on clinical and biomarker endpoints.^{e,22}

5. Risk Assessment & Safe-Use Conditions

The safety database includes the placebo-controlled pool, which is comprised of 984 patients in the clinical development program who received at least one dose of donanemab including 853 patients in the placebo-controlled period of AACI and 131 patients exposed to monotherapy with donanemab in study AACG.⁷ Supportive information was provided from phase 2 Study AACG, Study IST-MC-AACH Part B (TRAILBLAZER-EXT [NCT04640077]), long term extension period of Study AACI, Safety Addendum of study AACI, and Study AACN-Dona Cohort (TRAILBLAZWER-ALZ4 [NCT05108922]). Supportive data from the AACI long-term extension period and the open-label safety addendum (n=1692), and the two open-label studies, AACH Part B (n=55) and AACN Dona-Cohort (n=71), were pooled with donanemab-treated patients from Studies AACG and the placebo-controlled period of Study AACI to create an all-donanemab pool (n=2802).⁷

Regarding the relevant characteristics of the safety population, the clinical safety reviewer notes that the risk of using donanemab in patients <65 years of age or ≥85 years of age, with advanced comorbidities, or with more advanced states of AD could be higher as the population studied was relatively healthy, and had mild cognitive impairment or mild AD.⁷

The most common adverse reactions reported in at least 5% of donanemab treated patients include ARIA-H microhemorrhage, ARIA-E, ARIA H superficial siderosis, headache, and infusion related reactions.⁵

Hypersensitivity and infusion-related reactions as well as ARIA and intracranial hemorrhage are topics of special interest and are discussed in more depth below (see Section 5.2).

^e Section 505-1 (a) of the FD&C Act: *FDAAA factor (C): The expected benefit of the drug with respect to such disease or condition.*

5.1. Serious Adverse Events (SAE)^f

5.1.1. Deaths

In the primary safety database (placebo-controlled pool), there was a higher incidence of deaths reported in donanemab treated patients compared to placebo (1.7% [17/984] versus 0.8% [8/999], respectively). According to the clinical safety reviewer, the higher incidence of death was partially attributable to a higher incidence of fatal cases related to ARIA observed in the placebo-controlled pool (0.3% [3/984] in donanemab-treated participants versus none [0/999] on placebo).⁷

Overall, in the all-donanemab-pool, there were 32 deaths (1.1% [32/2802]) among donanemab treated patients as of the 90-Day Safety Update. Five deaths were related to either ARIA or intracranial hemorrhage. See Section X below for further details on ARIA or ICH related deaths.

Other causes of death include pneumonia (n=6), cerebral infarction/cerebral ischemia (n=2), dehydration (n=2), pulmonary embolism (n=1), respiratory failure (n=2), suicide (n=2), falls or trauma (n=3), acute myocardial infarction (n=1), renal failure (n=1), cardiac failure congestion (n=1), pancreatic carcinoma metastatic (n=1), retroperitoneal hemorrhage (n=1), metabolic encephalopathy (n=1), aortic dissection (n=1), and unknown (n=2). The clinical safety reviewer concluded that the role of donanemab was likely in 4 of the 6 cases of pneumonia given that pneumonia was among the SAEs with highest incidence in the donanemab treated patients.⁷ The subjects in all of the fatal cases of pneumonia had risk factors for pneumonia including age $\geq$ 75 years and dementia.⁷

5.2. Adverse Events of Special Interest (AESI)

5.2.1. Amyloid Related Imaging Abnormalities (ARIA)

Amyloid related imaging abnormalities (ARIA), a side effect known to occur with the class of monoclonal antibodies targeting beta amyloid, can take the form of edema (ARIA-E) and hemosiderin deposition (ARIA-H). Although rare, ARIA can occur spontaneously in patients with Alzheimer's disease particularly in patients with MRI findings suggestive of cerebral amyloid angiopathy, such as pretreatment microhemorrhage or superficial siderosis. The risk of ARIA-E is higher with the use of beta amyloid monoclonal antibodies, in particular with treatment initiation, in apolipoprotein e4 variant (ApoE ϵ 4) carriers, and with higher dosages.²³ ARIA-H can present on MRI as cerebral microhemorrhages, hemosiderosis, or cerebral hemorrhages.²³ Additionally, spontaneous microhemorrhages overall have a prevalence of ~5% in the elderly with higher prevalence in those with cerebrovascular risk factors.²⁴ ARIA usually occurs early in treatment with monoclonal antibodies targeting beta amyloid and is usually

^f Any adverse drug experience occurring at any dose that results in any of the following outcomes: Death, a life-threatening adverse drug experience, inpatient hospitalization, or prolongation of existing hospitalization, a persistent or significant disability/incapacity, or a congenital anomaly/birth defect. Important medical events that may not result in death, be life-threatening, or require hospitalization may be considered a serious adverse drug experience when, based upon appropriate medical judgment, they may jeopardize the patient or subject and may require medical or surgical intervention to prevent one of the outcomes listed in this definition.

asymptomatic.⁵ However, serious life-threatening events and fatalities have occurred.⁵ Symptomatic ARIA occurred in 6% of patients treated with donanemab with resolution of clinical symptoms occurring in 85% of patients.⁵ When ARIA symptoms are present, they may include, but are not limited to, headache, confusion, visual changes, dizziness, nausea, and gait difficulty.⁵ Focal neurologic deficits may also occur and they can mimic ischemic stroke.⁵

ARIA Monitoring

In Study AACG, ARIA monitoring with safety MRIs was conducted during screening, prior to the 2nd, 4th, 5th, 7th, 10th, 14th infusion, week 76, and the early discontinuation visit.²⁵ Similar frequency of safety MRIs were conducted in study AACI for ARIA monitoring including during screening, prior to the 2nd, 4th, 7th infusion, week 52, week 76, and the early discontinuation visit. Additional MRIs were conducted at unscheduled intervals if a patient developed more than one symptom suggestive of ARIA-E or a single symptom suggestive of ARIA-E of sufficient severity. If ARIA was present, then MRIs were repeated every 4 to 6 weeks until resolution of ARIA-E or stabilization of ARIA-H. The proposed labeling will include recommendations for monitoring with MRIs prior to initiating treatment, prior to the 2nd, 3rd, 4th, and 7th infusions and if a patient experiences symptoms suggestive of ARIA.

Dosing Modifications for Patients with ARIA

Dosing modification or discontinuation per protocol for the first occurrence of ARIA-E during study AACG was based on symptoms and radiologic severity of ARIA-E.²⁵ If a dose reduction was required, donanemab was reduced to the next lower dose such as 700 mg for a patient previously receiving 1400 mg dose or placebo for a patient previously receiving 700 mg dose) and an unscheduled safety MRI was required 4 to 6 weeks after restarting donanemab until ARIA-E was resolved. If ARIA-E symptoms and radiologic findings were not completely resolved within 16 weeks, donanemab treatment was permanently discontinued. No dose reductions were done for ARIA-H and donanemab was discontinued for patients with symptomatic ARIA-H; >4 new microhemorrhages, > 1 new area of superficial siderosis or significant worsening of pre-existing superficial siderosis; or any macrohemorrhage, regardless of symptoms. All new cases of ARIA-H required unscheduled MRI scans every 4-6 weeks until ARIA-H was stabilized without new findings.

Dosing modifications and discontinuations for ARIA-E and ARIA-H that occurred during study AACI were similar to dosing modifications and discontinuations for ARIA-E during study AACG. Dosing modification and discontinuations for ARIA-E and ARIA-H during study AACI were based on symptoms and radiologic severity. During studies AACG and AACI, donanemab was permanently discontinued for cases of intracerebral hemorrhage. If ARIA was present, then MRIs were repeated every 4 to 6 weeks until resolution of ARIA-E or stabilization of ARIA-H. The proposed labeling will include recommendations for dosing interruptions in patients with ARIA.

ARIA Incidence and Severity

A higher incidence of ARIA was observed in donanemab treated patients compared to placebo.^{6,7,25} In study AACG, ARIA (-E and/or -H) was observed in 37% (37/131) of patients treated with donanemab compared to 8% (10/125) of patients on placebo.²⁵ There were no deaths due to ARIA in study AACG; however, five deaths related to ARIA did occur in Study AACI (see below).⁷ In the placebo-controlled period of study AACI, ARIA (-E or -H) was observed in 36% (307/853) of patients treated with donanemab compared to 14% (122/874) of patients on placebo.⁵

In study AACI, the of ARIA-E occurred in 24% of donanemab treated patients compared to 2% of patients on placebo.⁵ The maximum radiographic severity of ARIA-E in patients treated with donanemab was mild in 7%, moderate in 15%, and severe in 2% of patients.⁵ ARIA-E radiographically resolved in 63% of patients treated with donanemab by 12 weeks and by 20 weeks in 80% of patients.⁵ Similar results were reported in study AACG.

In study AACI, ARIA-H was observed in 31% of patients treated with donanemab, compared to 13% of patients on placebo.⁵ The maximum radiographic severity of ARIA-H microhemorrhage in patients treated with donanemab was mild in 17%, moderate in 4%, and severe in 5% of patients.⁵ Similar results were reported in study AACG. The protocols for studies AACI and AACG did not require that ARIA-H be reported as an adverse event, rather captured as an MRI finding. Symptoms of ARIA-H were not required to be collected or graded as mild, moderate, and severe as an adverse event. According to the clinical safety reviewer, the method of collection of adverse events of ARIA-H likely results in under recording of ARIA-H events.⁷

Deaths due to ARIA

The clinical safety reviewer identified five fatal cases related to either ARIA-E, ARIA-H, or combined ARIA-E and -H. Two of the fatal cases (2/5) also had cerebral hemorrhage as described below.⁷ The clinical safety reviewer concluded that three of the five fatal ARIA cases (ARIA-E [n=2]; and ARIA-E and ARIA-H [n=1]) are related to donanemab and the fifth case (ARIA-H [n=1]) is likely related to donanemab.⁷ The clinical safety reviewer recommends that a boxed warning be added to the prescribing information (PI) indicating that fatalities in patients with ARIA have occurred. This boxed warning has also been included in the labeling of other amyloid-directed antibodies.⁷

ARIA in ApoE ε4 carriers

Similar to other anti-amyloid beta monoclonal antibodies, a higher incidence of ARIA-E and ARIA-H was observed in ApoE ε4 carriers treated with donanemab. In study AACI, severe radiographic ARIA-E occurred more frequently in patients treated with donanemab that were ApoE ε4 homozygotes (3% [4/143]) and heterozygotes (2% [9/452]) compared to noncarriers ((0.4% [1/255])).⁵ Radiographically severe ARIA-H in donanemab treated patients occurred more frequently in ApoE ε4 homozygotes (22%

⁶ Section 505-1 (a) of the FD&C Act: *FDAAA factor (E): The seriousness of any known or potential adverse events that may be related to the drug and the background incidence of such events in the population likely to use the drug.*

[31/143]) and heterozygotes (8% [38/452]) compared to noncarriers (4% [9/255]). Labeling is expected to convey the risk of ARIA in AD patients who are ApoE ϵ 4 homozygotes and include recommendations for testing for ApoE ϵ 4 status prior to initiation of treatment.⁵ While the boxed warning is expected to recommend testing for ApoE ϵ 4 status prior to initiation of treatment to inform the risk of developing ARIA, it is important to note that an FDA-authorized test for detection of ApoE ϵ 4 alleles is not currently available. Section 5 is expected to include recommendations that patients can still be treated with donanemab when genotype testing is not performed; however, without testing, it cannot be determined if the patient is ApoE ϵ 4 carrier.

Intracerebral Hemorrhage

There were four fatal cases related to intracranial hemorrhage identified in the donanemab pool. Two of the cases occurred in conjunction with ARIA findings. In the first case, a patient presenting with symptoms mimicking stroke who was administered tissue plasminogen activator (tPA) and developed multiple intracerebral hemorrhages. ARIA-E was noted on the post tPA MRI, and the patient subsequently died. The clinical safety reviewer concluded that the relationship of cerebral hemorrhage to donanemab is uncertain in the setting of treatment with tPA.⁷ The second case of intracerebral hemorrhage was observed concurrently in the patient with a fatality from ARIA-H as noted above. The clinical safety reviewer notes that in both of these cases related to ARIA and cerebral hemorrhage, the patients had risk factors for the development of cerebral hemorrhage including MRI findings of superficial siderosis and white matter disease that can be associated with cerebral amyloid angiopathy (CAA), use of ingestion of acetylsalicylic acid, presence of ApoE ϵ 4 allele.⁷

The other two additional fatal cases related to intracranial hemorrhage include one case of subarachnoid hemorrhage that preceded an event of fall from a ladder and one case of thalamus hemorrhage in a patient with a history of hypertension and increased blood pressure on day of event. Overall, the clinical safety reviewer concluded that of the four cases, the two cases of intracranial hemorrhage in the setting of ARIA are likely related to donanemab and that the role of donanemab in the two remaining cases could not be ruled out.⁷ Regarding the risk of cerebral hemorrhage, the clinical safety reviewer recommends including a statement regarding fatal cases of cerebral hemorrhage occurring in the setting of thrombolytic administration be included in the Warnings and Precautions section of the PI.⁷ The clinical safety reviewer also recommends that healthcare providers and patients be made aware that ARIA can present with focal neurologic symptoms that can mimic a stroke.⁷ Additional recommendations include that patients should carry a medication information card indicating that they are being treated with donanemab and that donanemab can cause ARIA with symptoms that may mimic ischemic stroke.⁷

In study AACI, intracerebral hemorrhage greater than 1 cm in diameter was reported in 0.5% (4/853) of donanemab treated patients compared to 0.2% (2/874) of patients on placebo.⁵ The incidence of intracerebral hemorrhage greater than 1 cm in diameter was 0.6% (2/349 patients) in donanemab treated patients on a concomitant antithrombotic medication compared to 0.4% (2/504) in those who did not receive an antithrombotic.⁵ The clinical safety reviewer concluded that the number of events and the limited exposure to non-aspirin antithrombotic medications limit definitive conclusions about the risk of ARIA or intracerebral hemorrhage in patients taking antithrombotic medications.⁷

Similar to other amyloid-directed antibodies, the clinical safety reviewer recommends communicating the risk of ARIA and the risk of intracerebral hemorrhage in a boxed warning including statements that fatal cases of ARIA have been reported, that serious intracerebral hemorrhages can occur, and that ARIA-E can cause focal neurologic deficits that can mimic ischemic stroke.⁷ Additionally, the boxed warning will include the risk of ARIA in patients who are ApoE ϵ 4 homozygotes.⁵ Guidance regarding monitoring and implications regarding a finding of ARIA on subsequent dosing will be conveyed in Section 2 of the prescribing information.⁵ Additional information regarding risk of ARIA and intracerebral hemorrhage in Section 5 includes radiographic classification criteria for ARIA severity, the need for assessment of symptoms associated with ARIA throughout treatment, and the need for testing for ApoE ϵ 4 status prior to initiation of treatment.⁵ The clinical safety reviewer notes uncertainties with the safety of concomitant use of medications that increase bleeding risk, safety of treating AD patients through episodes of cerebral hemorrhage greater than 1 cm, as well as the safety of treating patients otherwise at risk for bleeding, therefore, agrees with the applicant's proposed enhanced post marketing pharmacovigilance strategy to conduct observational safety studies to address hypersensitivity reactions and further characterize ARIA-E and ARIA-H in donanemab treated patients.^{7,25} In addition, the clinical safety reviewer recommends a registry-based, prospective, observational study to evaluate safety outcomes among Alzheimer's disease patients treated with donanemab, using, for example, the Alzheimer's Network for Treatment and Diagnostics (ALZ-NET) registry, including patients who are ApoE ϵ 4 homozygotes, and/or exposed to antithrombotics or thrombolytics, and/or have a diagnosis of, or imaging findings consistent with a high risk for, cerebral amyloid angiopathy.⁷

5.2.2. Hypersensitivity and Infusion Related Reactions

Hypersensitivity and infusion related reactions have occurred patients being treated with donanemab. In the placebo controlled period of Study AACI, infusion-related reactions were observed in 8.7% of patients treated with donanemab versus 0.5% on placebo.^{7, h} Symptoms associated with infusion-related reactions include chills, erythema, nausea/vomiting, dyspnea, sweating, elevated blood pressure, headache, chest pain, and low blood pressure. Non-infusion related hypersensitivity reaction including anaphylaxis and angioedema occurred in 7.3% of donanemab-treated patients vs 5.0% on placebo. Serious cases of infusion related reactions occurred at the time of the infusion or within minutes following the end of infusion. In two cases of serious infusion reactions, increases in blood pressure were associated with subarachnoid hemorrhage and myocardial infarction.

Fifteen (15) cases of anaphylaxis were identified by the clinical safety reviewer in the all-donanemab pool. The clinical safety reviewer determined 10 of those 15 reported cases as likely anaphylactic reactions.⁷ Reported interventions for infusion related reactions included administration of medications prior to infusion and slowing of the infusion on or after the infusion related reaction. Similar incidences of infusion related reactions were observed in subjects who received pretreatment medication upon

^h Section 505-1 (a) of the FD&C Act: *FDAAA factor (E): The seriousness of any known or potential adverse events that may be related to the drug and the background incidence of such events in the population likely to use the drug.*

rechallenge (38.1%, 16/42) compared to those who did not receive pretreatment medication on rechallenge (41.0%, 57/139).⁷ However, according to the clinical safety reviewer, the number of participants receiving interventions were too few to draw conclusions.⁷

The clinical safety reviewer recommends including the risk of infusion reactions and hypersensitivity in Section 5, warnings and precautions section of the label and in the Medication Guide.⁷

6. Expected Postmarket Use

The anticipated prescribing population for donanemab will likely be memory disorder specialists which include neurologists, psychiatrists, and geriatricians, who should have experience in the care for patients with AD. Donanemab will likely be administered in settings with medical personnel equipped to handle adverse events, such as infusion centers and provider clinics/offices. As donanemab will require brain MRI evaluation prior to treatment initiation and during therapy, radiologists will be involved in the care of patient administered donanemab to identify ARIA and classify radiographic severity. Memory disorder specialists and neuroradiologists may be aware of the risk of ARIA since ARIA has been observed in clinical trials with other anti-amyloid beta monoclonal antibodies including aducanumab and lecanemab.

7. Risk Management Activities Proposed by the Applicant

The Applicant did not propose any risk management activities for donanemab beyond routine pharmacovigilance and labeling. However, the Applicant proposed a voluntary healthcare professional educational program related to ARIA identification and management.²⁶ The Applicant also intends to voluntarily provide a safety card for patients [REDACTED] (b) (4), and that providers should be aware the patient is taking donanemab.²⁷

8. Discussion of Need for a REMS

The review team recommends approval of donanemab on the basis of the efficacy and safety information currently available.²²

AD is a progressive, degenerative neurologic disorder characterized by progressive memory loss, behavioral problems, and inability to perform activities of daily living.¹³ AD is the most common cause of dementia and a leading cause of morbidity and mortality in the aging population. Most options target symptoms of AD. The first two FDA approved treatments that target the underlying pathophysiology associated with AD include Aduhelm (accelerated approval) and Leqembi (traditional approval).

As there are no approved treatments designed to cure the disease, there remains an unmet need for treatment options for AD. Donanemab offers an additional anti-amyloid beta MAb option for the treatment of AD. In Study AACI, the use of donanemab in AD resulted in statistically significant reduction on the DN1's recommended endpoint to demonstrate clinical decline, CDR-SB.

A Peripheral and Central Nervous System (PCNS) Advisory Committee meeting was held on June 10, 2024 to discuss whether the data from the Phase 3 Study AACI (TRAILBLAZER-ALZ 2) support a favorable

benefit-risk assessment for the use of donanemab for the treatment of Alzheimer's disease (AD).¹² The eleven committee members unanimously voted "Yes" that the available data show that donanemab is effective for the treatment of Alzheimer's disease in the population enrolled in the clinical trials with mild cognitive impairment and mild dementia. In addition, eleven committee members unanimously voted "Yes" that the benefits outweigh the risks of donanemab in the treatment of AD in the population enrolled in the clinical trials with mild cognitive impairment and mild dementia. A REMS proposal was not discussed.

Donanemab is associated with increased risk of ARIA (-E and/or -H), which has also been observed in clinical development programs for aducanumab, lecanemab, as well as other investigational anti-amyloid antibodies. The majority of ARIA is asymptomatic and usually mild or moderate in severity. In most cases, ARIA resolved radiographically and symptomatically over time after detection. ARIA can be detected by MRI and managed with close MRI monitoring and dose adjustments. Five cases of ARIA in study AACI had fatal outcomes. Two of the cases involved intracerebral hemorrhage in patients who had risk factors for cerebral hemorrhage. One of the fatal intracerebral hemorrhage cases involved a patient with ARIA-E, with symptoms mimicking stroke, who was treated with thrombolytic therapy.

Prescribers are likely to be familiar with the risk of ARIA, which is well documented for the approved anti-amyloid beta monoclonal antibodies, aducanumab and lecanemab.^{19,28} The risk of ARIA is also described in published expert recommendations on appropriate use for aducanumab.^{23,29} Additionally, the risk of ARIA, including a statement regarding fatal cases of intracerebral hemorrhage occurring in the setting of thrombolytic administration, and a statement regarding ARIA-E symptoms that can mimic ischemic stroke will be communicated through a boxed warning, Section 5: Warnings and Precautions, and in the Medication Guide.⁵ At the time of this review, final labeling and post marketing requirements (PMR) were still under negotiation. Labeling is expected to convey the risk of ARIA and include recommendations for MRI monitoring, radiographic classification criteria for ARIA severity, the need for assessment of symptoms associated with ARIA throughout treatment, considerations for continuing donanemab in the setting of ARIA, and testing for ApoE ε4 status prior to initiation of treatment. The Applicant also intends to provide a safety card for patients that includes ARIA-related information including details to inform and aid healthcare providers in managing emergency situations.²⁶ The clinical safety reviewer agrees with the Applicants proposal of enhanced post-market pharmacovigilance for the risk of ARIA and recommends additional PMR to conduct a registry-based, prospective, observational study as there should be additional data on the safety of donanemab in a real-world setting.⁷

Based on the data available, the seriousness of the disease state, and the prescribing community's likely familiarity with the risks associated with donanemab that do not pose unique REMS considerations compared with the risks associated with other therapies, this reviewer has concluded that should donanemab be approved, a REMS is not necessary to ensure the benefits outweigh the risks of donanemab. If new safety information becomes available, DRM can re-evaluate the need for a REMS.

9. Conclusion & Recommendations

Based on the available data a REMS is not necessary to ensure the benefits outweigh the risks of donanemab. The side effect profile is similar to other anti-amyloid beta monoclonal antibodies including the risk of ARIA and prescribers are likely to be familiar with and able to appropriately monitor for the risks. At the time of this review, the labeling and PMR negotiations were ongoing. Please notify DRM if new safety information becomes available that changes the benefit-risk profile; this recommendation can be reevaluated.

Should the Division of Neurology 1 have any concerns or questions or if new safety information becomes available, please send a consult to DRM.

10. Appendices

10.1. References

1. Eli Lilly and Company. Summary of Clinical Efficacy - donanemab-azbt. *BLA 761248*. May 18, 2022.
2. Eli Lilly and Company. BLA Submission - Kisunla (donanemab-azbt). *BLA 761248*. May 18, 2022 (sequence 002).
3. Dunn B. Complete Response Letter - Kisunla (donanemab-azbt). *BLA 761248*. January 18, 2023.
4. Eli Lilly and Company. BLA Resubmission - Kisunla (donanemab-azbt). *BLA 761248*. June 12, 2023 (sequence 104).
5. Eli Lilly and Company. DRAFT Kisunla (donanemab-azbt) Prescribing Information - as edited by FDA. *NDA 216718*. February 14, 2024.
6. Eli Lilly and Company. Clinical Overview - donanemab-azbt. May 18, 2022.
7. Branagan N. Division of Neurology 1. DRAFT Clinical Safety Review - Kisunla (donanemab-azbt). *BLA 761248*. June 25, 2024.
8. Eli Lilly and Company. Response To Information Request - Kisunla (donanemab-azbt). *BLA 761248*. July 31, 2023 (sequence 109).
9. Eli Lilly and Company. Response To Information Request - Kisunla (donanemab-azbt). *BLA 761248*. July 31, 2023 (sequence 112).
10. Eli Lilly and Company. Response To Information Request - Kisunla (donanemab-azbt). *BLA 761248*. September 5, 2023 (sequence 126).
11. Eli Lilly and Company. Response To Information Request - Kisunla (donanemab-azbt). *BLA 761248*. October 10, 2023 (sequence 139).
12. Food and Drug Administration. Peripheral and Central Nervous System (PCNS) Advisory Committee Meeting. FDA Briefing Document - donanemab-azbt. *BLA 761248*. June 10, 2024. <https://www.fda.gov/media/179166/download>
13. Keene CD, Montine TJ, Kuller LH. Epidemiology, pathology, and pathogenesis of Alzheimer Disease. In: DeKosky S, Wilterdink J, eds. *UptoDate*. 2022. Accessed January 4, 2023.
14. Alzheimer's Association. 2024 Alzheimer's Disease Facts and Figures. *Alzheimers Dementia*. 2024;20(5). Accessed June 27, 2024. <https://www.alz.org/media/documents/alzheimers-facts-and-figures.pdf>

15. Kochanek KD, Murphy SL, Xu J, Arias E. Mortality in the United States, 2022. *NCHS data brief*. 2024(492):1-8. Accessed June 27, 2024. chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/<https://www.cdc.gov/nchs/data/databriefs/db492.pdf>
16. Ganguli M, Dodge HH, Shen C, Pandav RS, DeKosky ST. Alzheimer Disease and Mortality: A 15-Year Epidemiological Study. *Archives of neurology (Chicago)*. 2005;62(5):779-784. doi:10.1001/archneur.62.5.779
17. Peripheral and Central Nervous System (PCNS) Drugs Advisory Committee Meeting. Combined FDA and Applicant PCNS Drugs Advisory Committee Briefing Document - aducanumab-avwa. *BLA 761178*. November 6, 2020. <https://public4.pagefreezer.com/content/FDA/04-03-2022T19:30/https://www.fda.gov/media/143502/download>
18. Press D, Buss S. Treatment of Alzheimer disease. In: Dekosky S, Schmader K, Yaffe K, eds. *UpToDate*. 2021
19. Eisai Inc. Leqembi (lecanemab-irmb) Prescribing Information. 07/2023. Accessed November 21, 2023. https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/761269Orig1s001lbl.pdf
20. Cavazzoni P. FDA's Decision to Approve New Treatment for Alzheimer's Disease - Aduhelm (aducanumab-avwa). *BLA 761178*. June 7, 2021. Accessed May 25, 2022. <https://www.fda.gov/drugs/news-events-human-drugs/fdas-decision-approve-new-treatment-alzheimers-disease>
21. Buracchio T. BLA Supplemental Approval Letter - Leqembi (lecanemab-irmb). *BLA 761269*. July 6, 2023. Accessed June 27, 2024. https://www.accessdata.fda.gov/drugsatfda_docs/appletter/2023/761269Orig1s001ltr.pdf
22. Krudys K. Division of Neurology 1. DRAFT Clinical Efficacy Review - Kisunla (donanemab-azbt). *BLA 761248*. June 26, 2024.
23. Withington CG, Turner RS. Amyloid-Related Imaging Abnormalities With Anti-amyloid Antibodies for the Treatment of Dementia Due to Alzheimer's Disease. *Frontiers in neurology*. 2022;13:862369-862369. doi:10.3389/fneur.2022.862369
24. Viswanathan A, Chabriat H. Cerebral microhemorrhage. *Stroke (1970)*. 2006;37(2):550-555. doi:10.1161/01.STR.0000199847.96188.12
25. Branagan N. Division of Neurology 1. Clinical Safety Review - Kisunla (donanemab-azbt). *BLA 761248*. January 18, 2023.
26. Eli Lilly and Company. Peripheral Central Nervous System (PCNS) Advisory Committee Meeting. Eli Lilly Briefing Document - donanemab-azbt. *BLA 761248*. June 10, 2024. <https://www.fda.gov/media/179167/download>
27. Eli Lilly and Company. Response To Information Request - Kisunla (donanemab-azbt). *BLA 761248*. March 1, 2024 (sequence 167).
28. Biogen Inc. Aduhelm (aducanumab-avwa) Prescribing Information. 08/2023. Accessed November 21, 2023. https://www.accessdata.fda.gov/drugsatfda_docs/label/2023/761178s011lbl.pdf
29. Cummings J, Rabinovici GD, Atri A, et al. Aducanumab: Appropriate Use Recommendations Update. *The journal of prevention of Alzheimer's disease*. 2022;9(2):221-230. doi:10.14283/jpad.2022.34

10.2. Table 1. Drugs Approved in the US for Alzheimer's Disease

Name (generic); Approval Year	Indication	Formulation(s)	Safety and Tolerability Issues	Risk Management Approaches
Aricept; Aricept ODT (donepezil hydrochloride); 1996	Treatment of dementia of the Alzheimer's type; efficacy has been demonstrated in patients with mild, moderate, and severe Alzheimer's disease	Oral tablets, oral disintegrating tablets	Safety issues related to increased cholinergic activity including the following: may exaggerate the effects of succinylcholine-type muscle relaxation during anesthesia; cardiac effects (bradycardia, heart block, syncopal episodes); peptic ulcer disease and gastrointestinal (GI) bleeding; nausea and vomiting; weight loss; bladder outflow obstruction; seizures; respiratory adverse events (caution in patients with pulmonary conditions)	Labeling – Warning and Precaution
Exelon (rivastigmine tartrate; rivastigmine transdermal system); 2000	Mild, moderate, and severe dementia of the Alzheimer's type and mild-to-moderate dementia associated with Parkinson's disease (PD)	Oral capsules; oral solution; transdermal patch	Significant gastrointestinal adverse reactions (nausea, vomiting, decreased appetite, weight loss, dehydration); allergic dermatitis; risks due to increased cholinergic activity (same as donepezil above); impairment in driving or use of machinery Patch also includes hospitalization and rarely death due to application of multiple patches at the same time; skin application site reactions	Labeling – Warning and Precaution
Razadyne; Razadyne ER (galantamine hydrobromide); 2001	Treatment of mild to moderate dementia of the Alzheimer's type	Oral immediate release and extended-release capsules; oral solution	Serious skin reactions (Stevens Johnson syndrome and acute generalized exanthematous pustulosis), risks due to increased cholinergic activity (same as donepezil above); deaths in subjects with mild cognitive impairment in two randomized trials	Labeling – Warning and Precaution

Namenda, Namenda XR (memantine hydrochloride); 2003	Treatment of moderate to severe dementia of the Alzheimer's type	Oral tablets; oral extended-release capsules	Genitourinary conditions (decreased urinary elimination of memantine with conditions that increase urinary pH)	Labeling – Warning and Precaution
Namzaric (memantine and donepezil hydrochloride); 2014	Treatment of moderate to severe dementia of the Alzheimer's type in patients stabilized on 10 mg of donepezil hydrochloride once daily	Oral extended-release capsules	Labeled safety issues are outlined for donepezil and memantine above	Labeling – Warning and Precautions
Aduhelm (aducanumab- avwa); 2021	Treatment of Alzheimer's disease; should be initiated in patients with mild cognitive impairment or mild dementia stage of disease, the population in which treatment was initiated in clinical trials	Intravenous infusion	Amyloid related imaging abnormalities-edema (ARIA-E), which can be observed on MRI as brain edema or sulcal effusions, and amyloid related imaging abnormalities-hemosiderin deposition (ARIA-H), which includes microhemorrhage and superficial siderosis	Labeling – Box Warning, Warning and Precautions
Leqembi (lecanemab- irmb); 2023	Treatment of Alzheimer's disease; should be initiated in patients with mild cognitive impairment or mild dementia stage of disease, the population in which treatment was initiated in clinical trials	Intravenous infusion	Amyloid related imaging abnormalities-edema (ARIA-E), which can be observed on MRI as brain edema or sulcal effusions, and amyloid related imaging abnormalities-hemosiderin deposition (ARIA-H), which includes microhemorrhage and superficial siderosis	Labeling – Box Warning, Warning and Precautions

Source: Information obtained from labeling from Drugs@FDA

This is a representation of an electronic record that was signed electronically. Following this are manifestations of any and all electronic signatures for this electronic record.

/s/

CARLA C DARLING
06/28/2024 03:08:08 PM

JACQUELINE E SHEPPARD
06/28/2024 03:16:33 PM

CYNTHIA L LACIVITA
06/28/2024 03:19:00 PM

Division of Risk Management (DRM)
Office of Medication Error Prevention and Risk Management (OMEPRM)
Office of Surveillance and Epidemiology (OSE)
Center for Drug Evaluation and Research (CDER)

Application Type	BLA
Application Number	761248
PDUFA Goal Date	January 18, 2023
Nexus TTT #	2022-2424
Reviewer Name(s)	Carla Darling, PharmD, BCPS
Team Leader	Jacqueline Sheppard, PharmD
Division Director	Cynthia LaCivita, PharmD
Review Completion Date	January 17, 2023
Subject	Deferral of Risk Evaluation and Mitigation Strategy (REMS) Review
Established Name	Donanemab-azbt
Trade Name	Kisunla
Name of Applicant	Eli Lilly and Company
Therapeutic Class	Anti-Amyloid Beta Monoclonal Antibody
Formulation(s)	Solution for Injection
Dosing Regimen	700 mg administered as an intravenous infusion every four weeks for the first three doses, followed by 1400 mg every four weeks

This memorandum (memo) is to defer the Division of Risk Management (DRM) review of the need for a Risk Evaluation and Mitigation Strategy (REMS) for Kisunla (donanemab-azbt), Biologic Licensing Application (BLA) 761248.

Eli Lilly and Company (Applicant) submitted the final portion of rolling BLA 761248 on May 18, 2022 with the proposed indication [REDACTED] ^{(b) (4)} of Alzheimer's disease (AD).¹ Donanemab, a new molecular entity (NME), is a recombinant human immunoglobulin gamma 1 (IgG1) monoclonal antibody (mAb) against aggregated insoluble plaque-bound amyloid beta. It is proposed as an intravenous infusion of 700 mg administered every four weeks for the first three doses, followed by 1400 mg every four weeks.² This application is under review in the Division of Neurology 1. The applicant did not submit a proposed REMS or risk management plan with this application

The efficacy and safety of donanemab was evaluated in one pivotal phase 2 trial. The study evaluated donanemab in patients with early AD defined both clinically (mild cognitive impairment AD and mild AD dementia) and by biomarkers (radiographic amyloid plaque levels plus low-to-medium tau levels). The primary endpoint was change in the composite measure integrated Alzheimer's Disease Rating Scale (iADRS) from baseline score to 76 weeks. During the review of the application, the review team identified issues regarding the size of safety database, which raised concerns regarding the long-term safety of donanemab. The clinical reviewer concluded that the safety database is insufficient to adequately characterize the long-term safety of donanemab in the treatment of Alzheimer's disease.³ Therefore, the Division of Neurology 1 plans to issue a Complete Response (CR) letter citing the inadequate safety database.

Therefore, DRM defers comment on the need for a REMS at this time. A discussion on the appropriate risk management strategy will be undertaken after the sponsor submits a satisfactory response to the Complete Response letter. Please send DRM a new consult request at such time. This memo serves to close the existing assignment for Kisunla (donanemab-azbt), BLA 761248.

References

1. Eli Lilly and Company. Clinical Overview for donanemab-azbt. May 18, 2022;
2. Eli Lilly and Company. DRAFT Kisunla (donanemab-azbt) Prescribing Information. *BLA 761248*. May 18, 2022;
3. Branagan N. Division of Neurology 1. DRAFT Clinical Safety Review for Donanemab-azbt. *BLA 761248*. January 17, 2023;

This is a representation of an electronic record that was signed electronically. Following this are manifestations of any and all electronic signatures for this electronic record.

/s/

CARLA C DARLING
01/17/2023 02:01:42 PM

JACQUELINE E SHEPPARD
01/17/2023 02:03:11 PM

CYNTHIA L LACIVITA
01/17/2023 02:30:13 PM